How are the climate and the economy linked?

ECON-4210
Economics of Climate Change
Class 3

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

1. Questions?
2. Attendance
3. Today's lecture:
    - **Linking the economy and the climate**
4. For next class…

Relationship between GHG emissions and economic output

- Stylized circular flow:
  - growth → emissions → warming → damages → policy → growth.
- Does climate change affect economic output directly?
- Do climate policies necessarily slow growth? Relative to what?

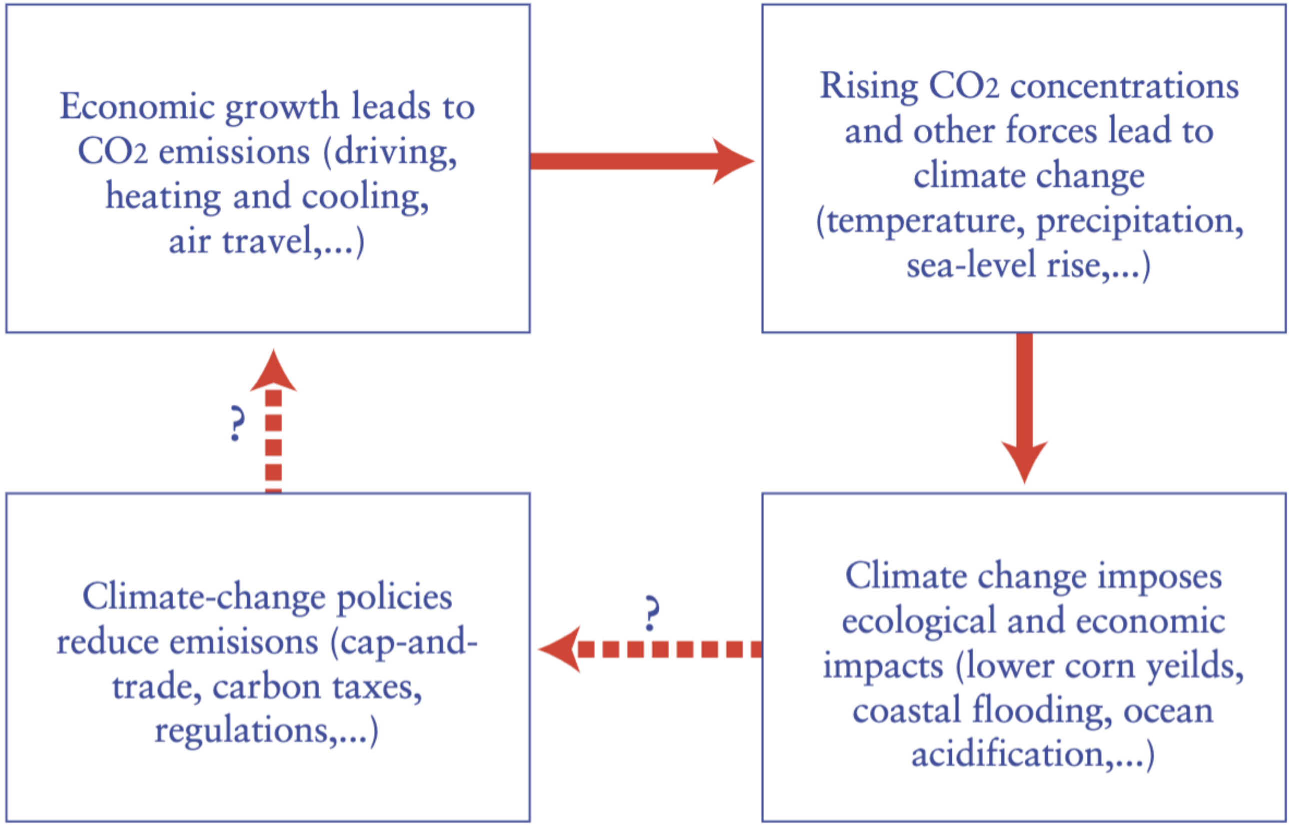
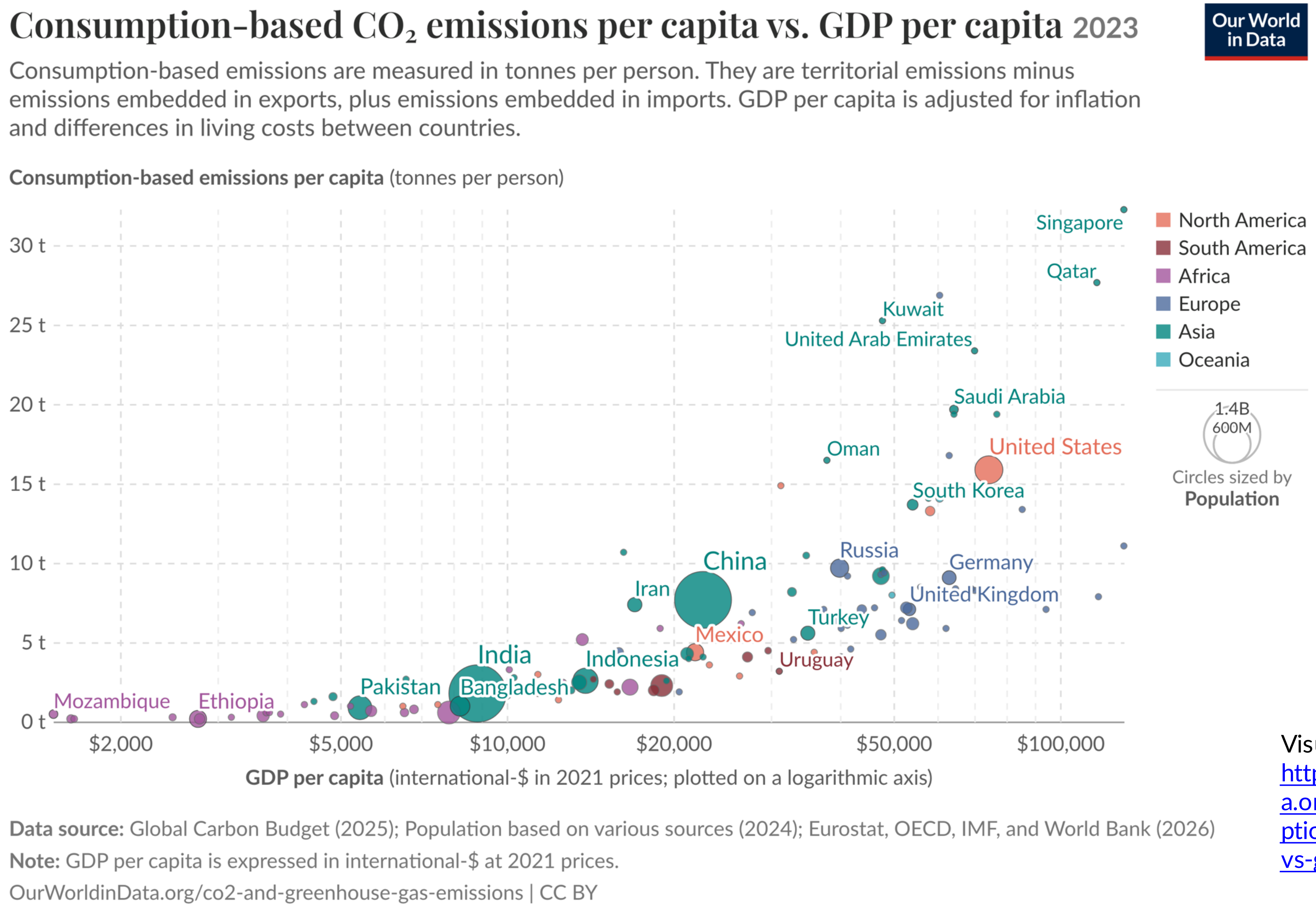


Figure 1. The circular flow of global warming science, impacts, and policy.

Source: Nordhaus (2013)

Richer countries emit more CO2 per person



Visualization: https://ourworldindata.org/grapher/consumption-co2-per-capita-vs-gdppc?time=2023

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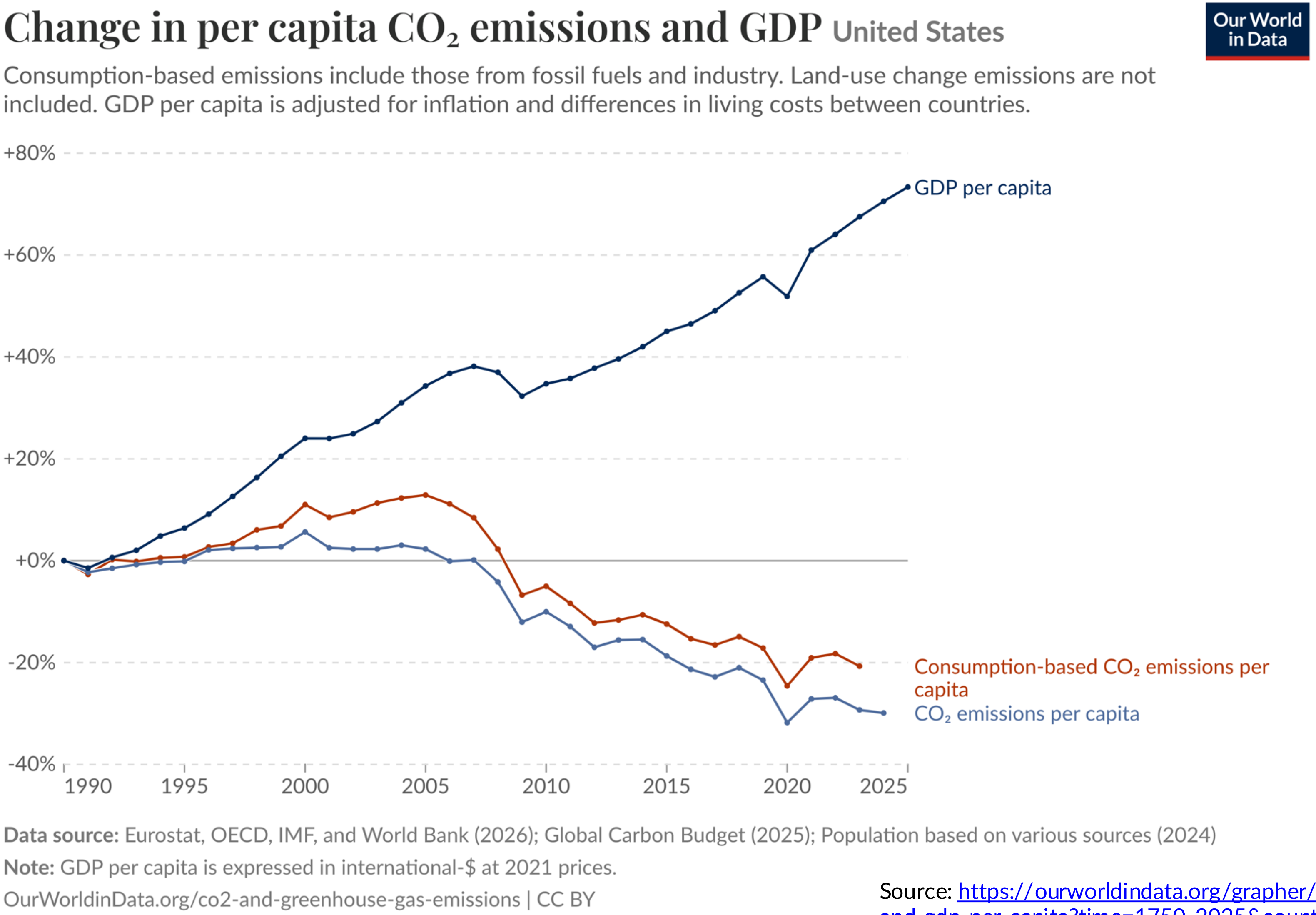
…but, economic growth has decoupled from emissions in wealthy countries



Source: https://ourworldindata.org/grapher/co2-emissions-and-gdp-per-capita?time=1750..2025&country=~USA

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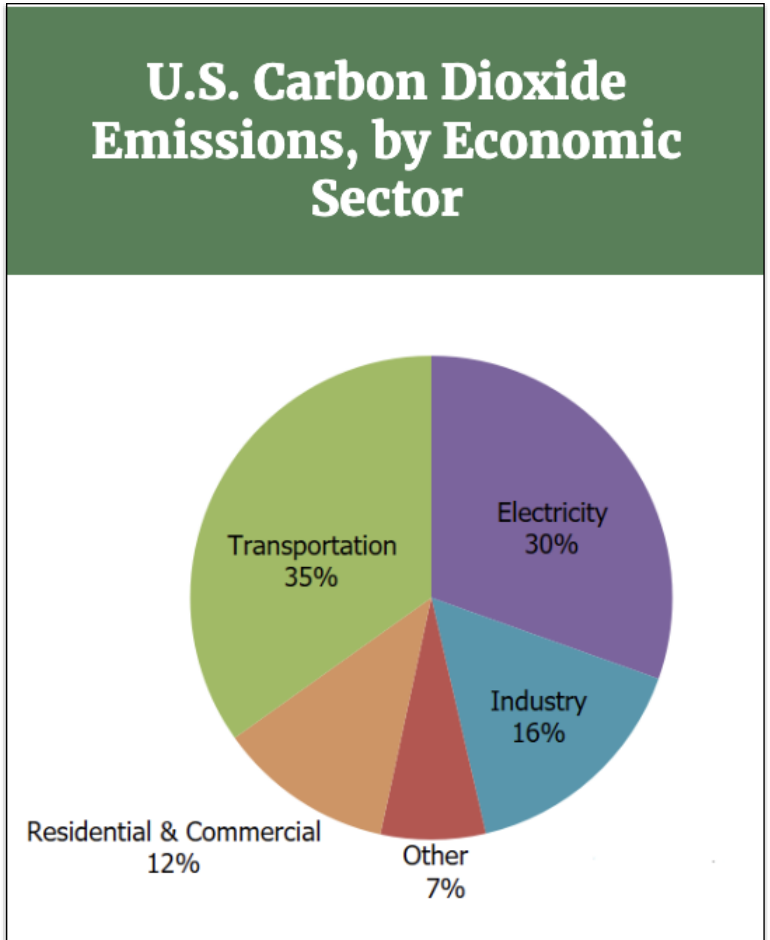
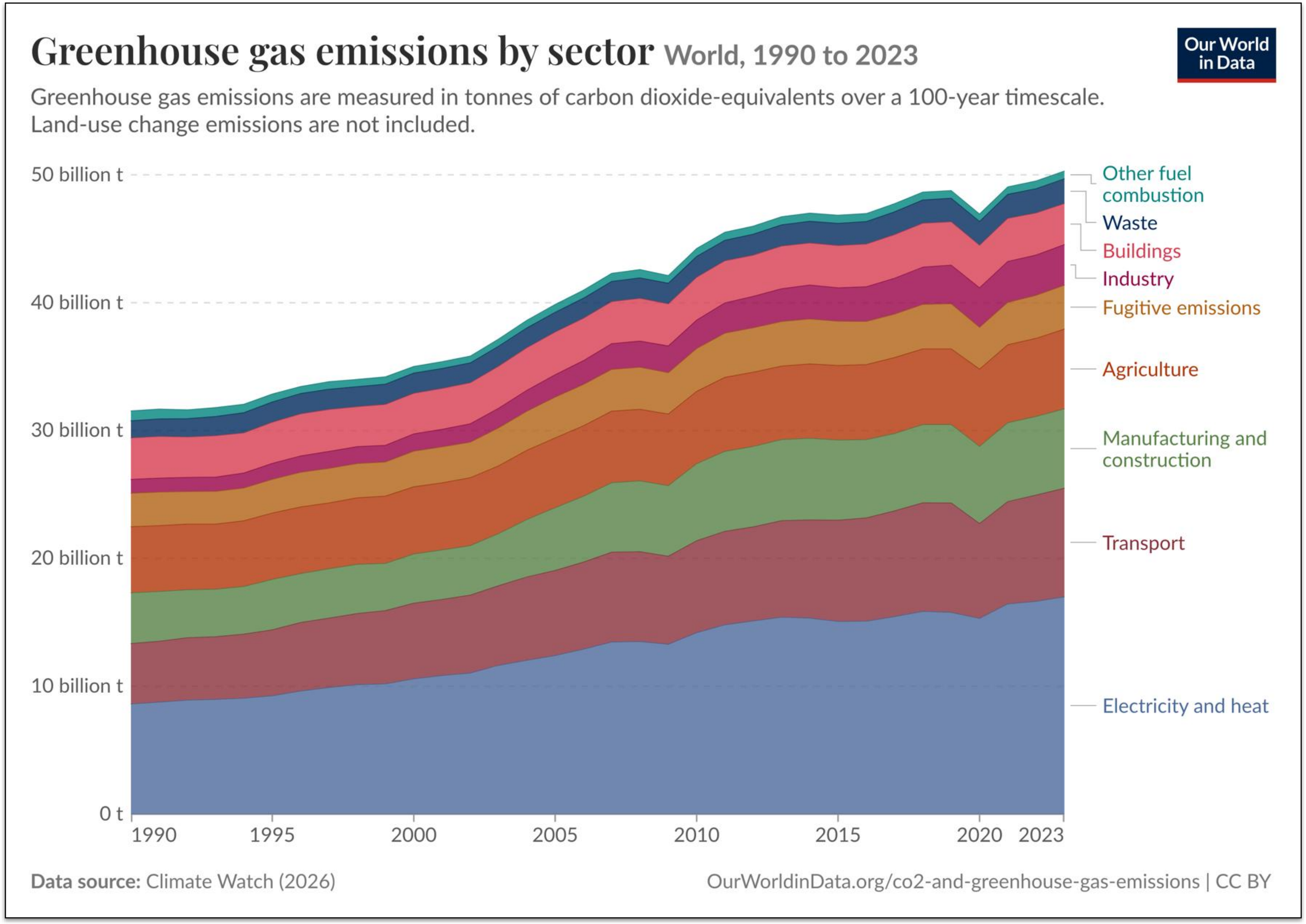
What sectors emit the most GHGs globally? …in the US?

Sources of GHG emissions…



Note: Total Emissions in 2022 are 6,343 Million Metric Tons of CO2 equivalent. Percentages may not add up to 100% due to independent rounding. Greenhouse gas



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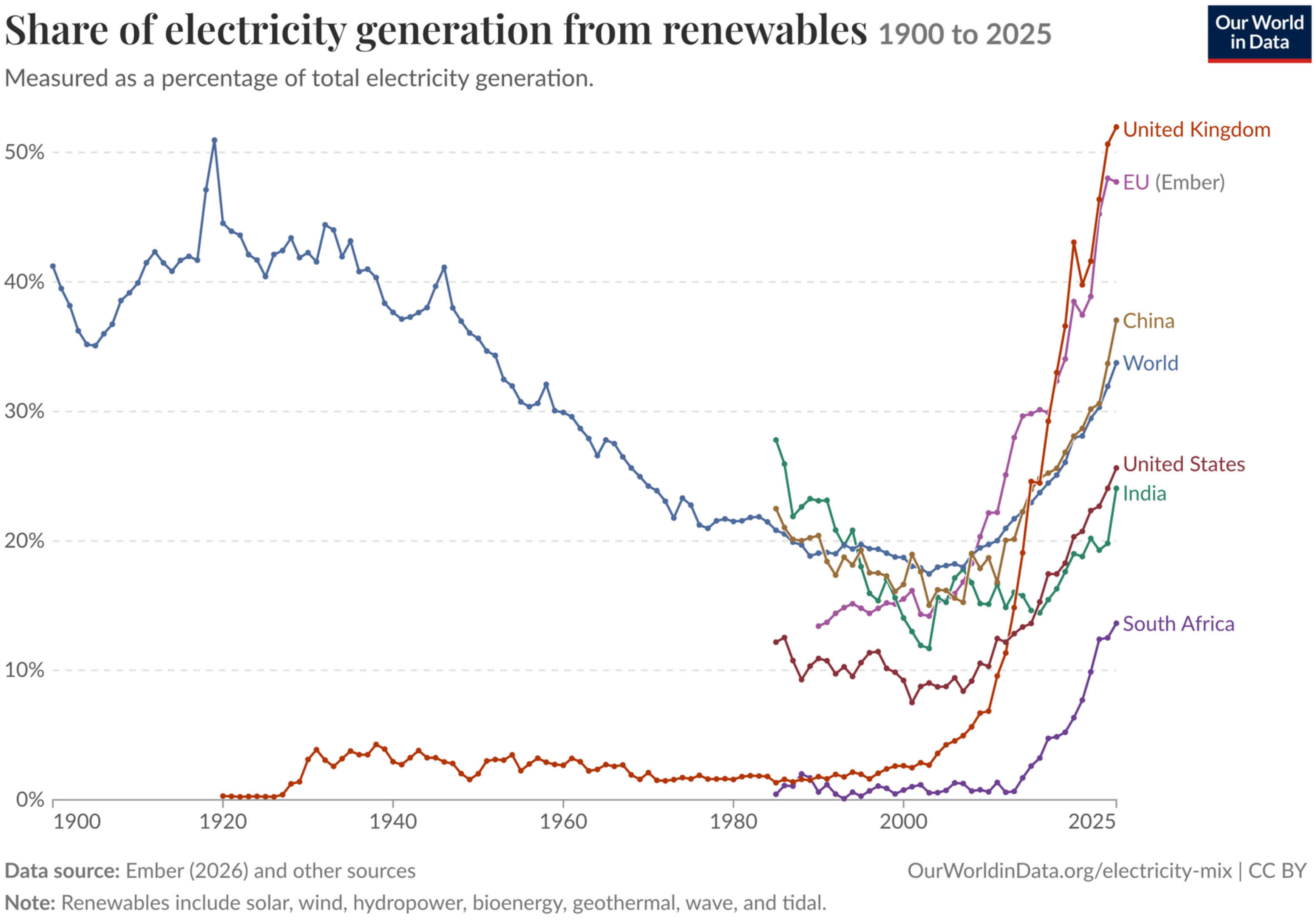
What percentage of electricity is generated from renewable sources (hydro, wind, solar, etc.) globally?
…in the US?

What percentage of electricity is generated from renewable sources (hydro, wind, solar, etc.) globally?
…in the US?

~34% globally
~26% in US



Source: https://ourworldindata.org/grapher/electricity-mix?tab=line&source=total&metric=generation&frequency=annual

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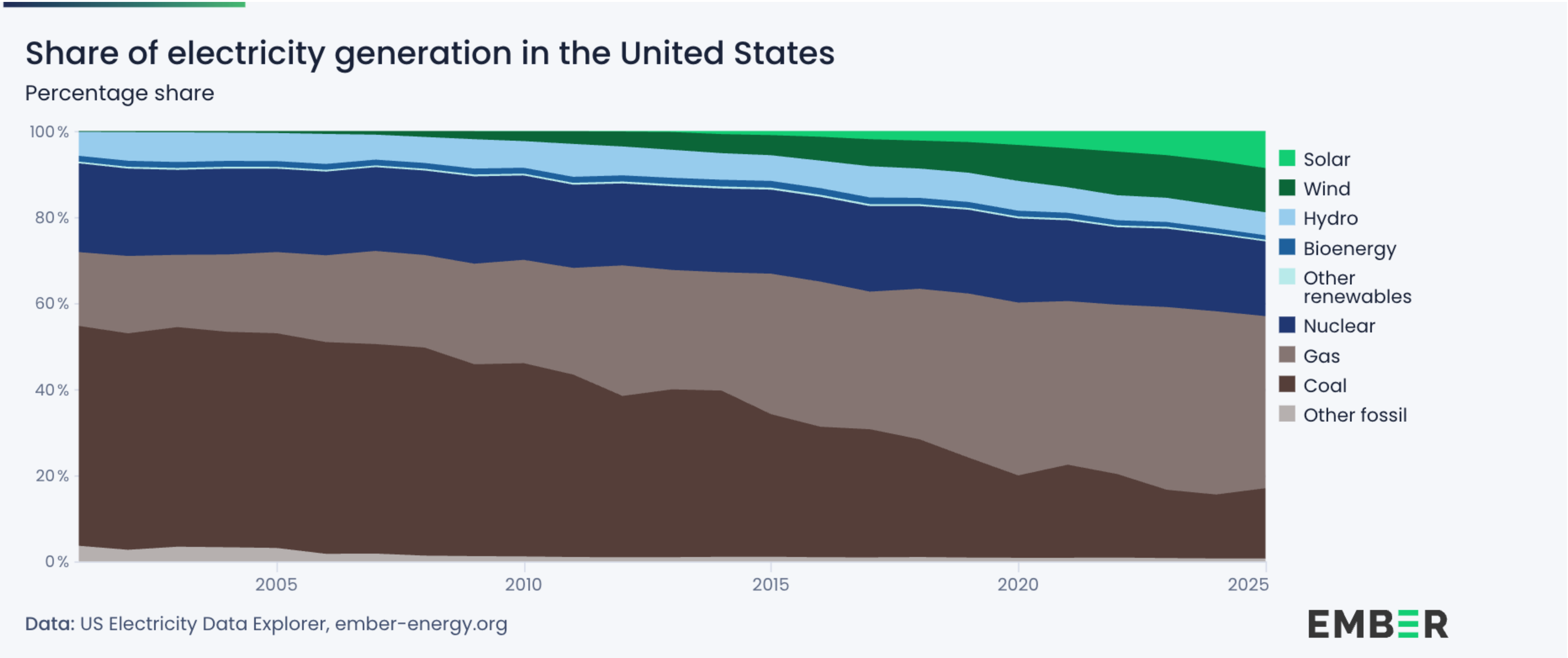
Fuel sources for electricity generation in US



Source: https://ember-energy.org/data/electricity-data-explorer/?entity=United+States&metric=pct_share

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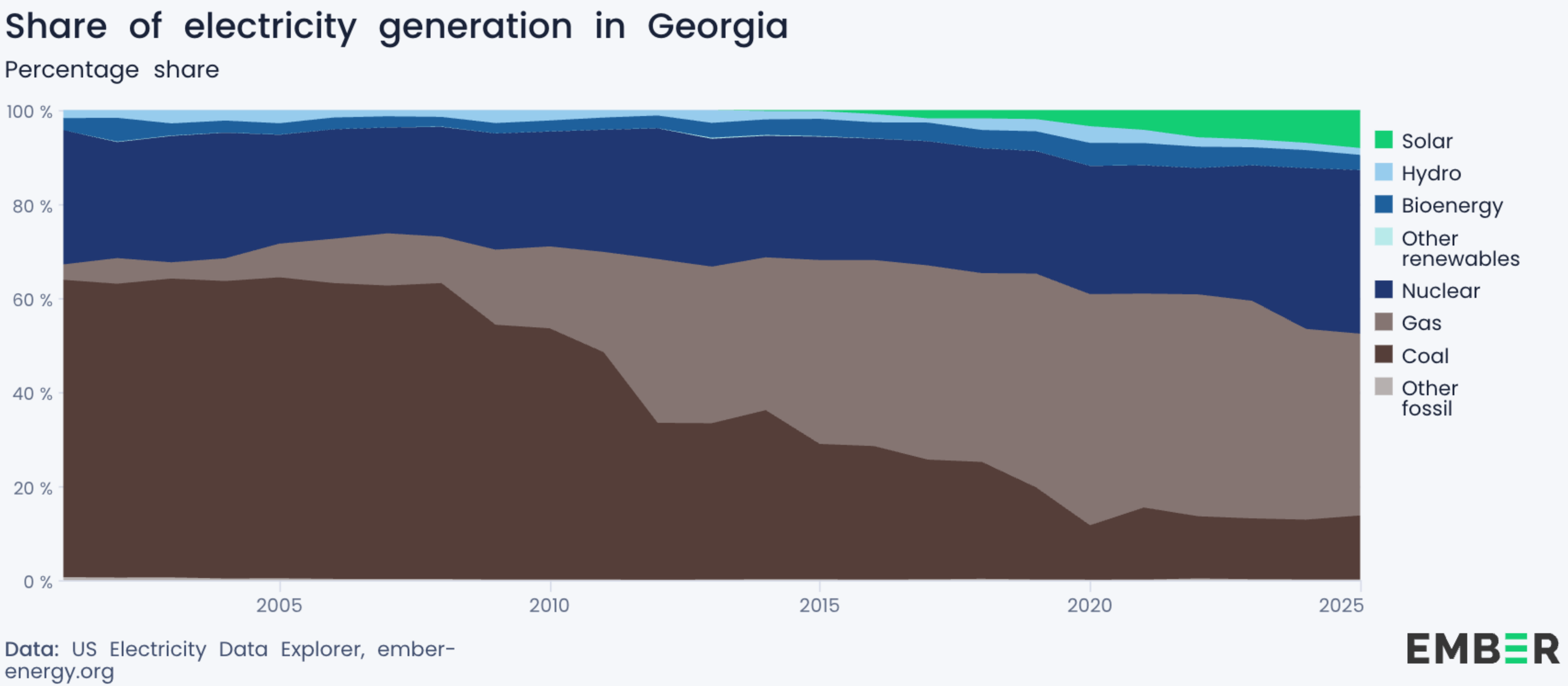
Fuel sources for electricity generation in GA



Fossil fuels have different carbon contents

Pounds of CO_2 emitted per million British thermal units (Btu) of energy for various fuels:	
Coal (bituminous)	205.7
Natural gas	117.0

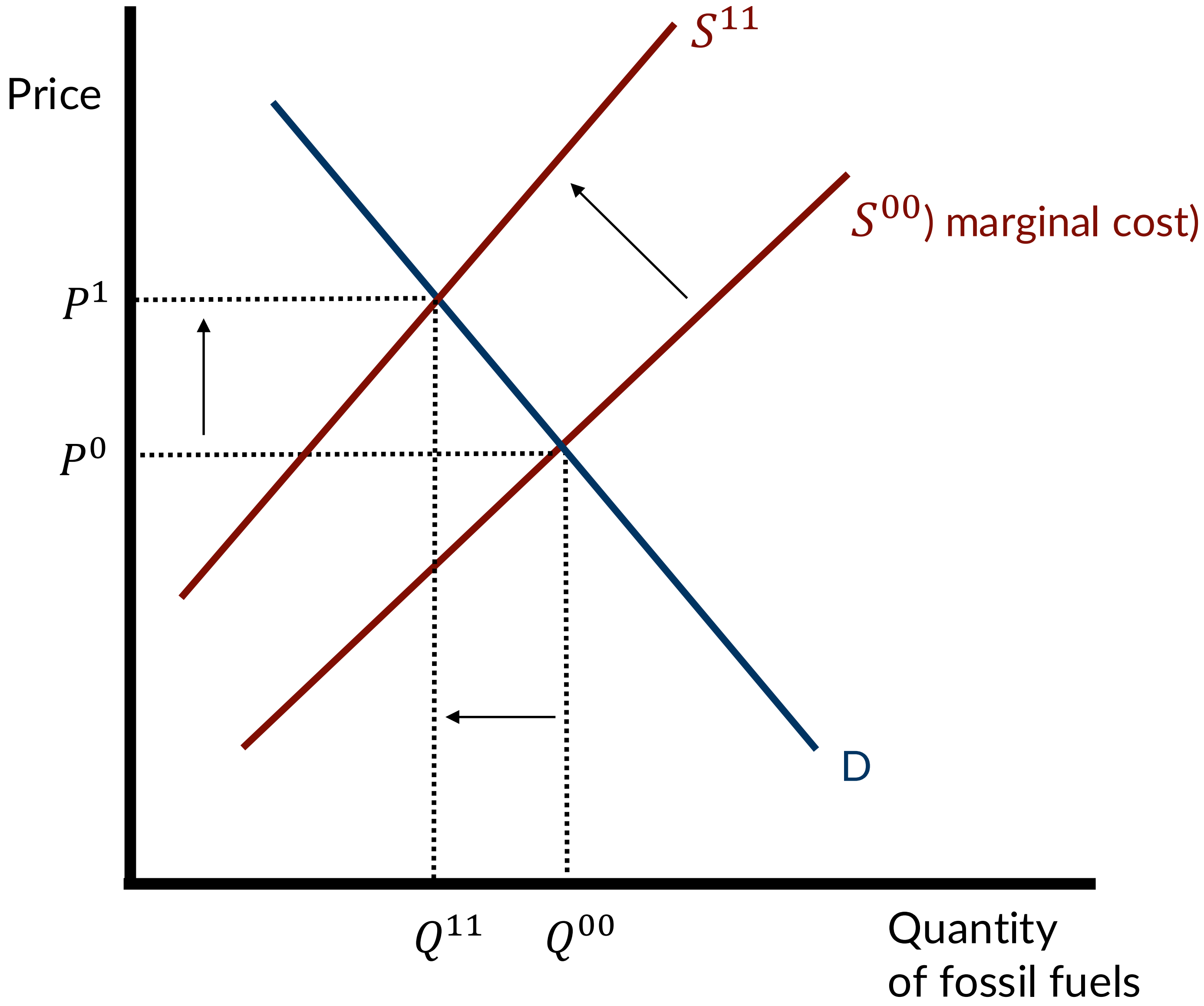
- Natural gas emits about 57% of the CO_2 that coal does per Btu.
- Cheap gas displaced coal in US generation after ~2008, largely due to hydraulic fracturing, which cut CO_2 emissions without any climate policy.

Will we ever stop using fossil fuels?

Covert, Greenstone, & Knittel (2016)

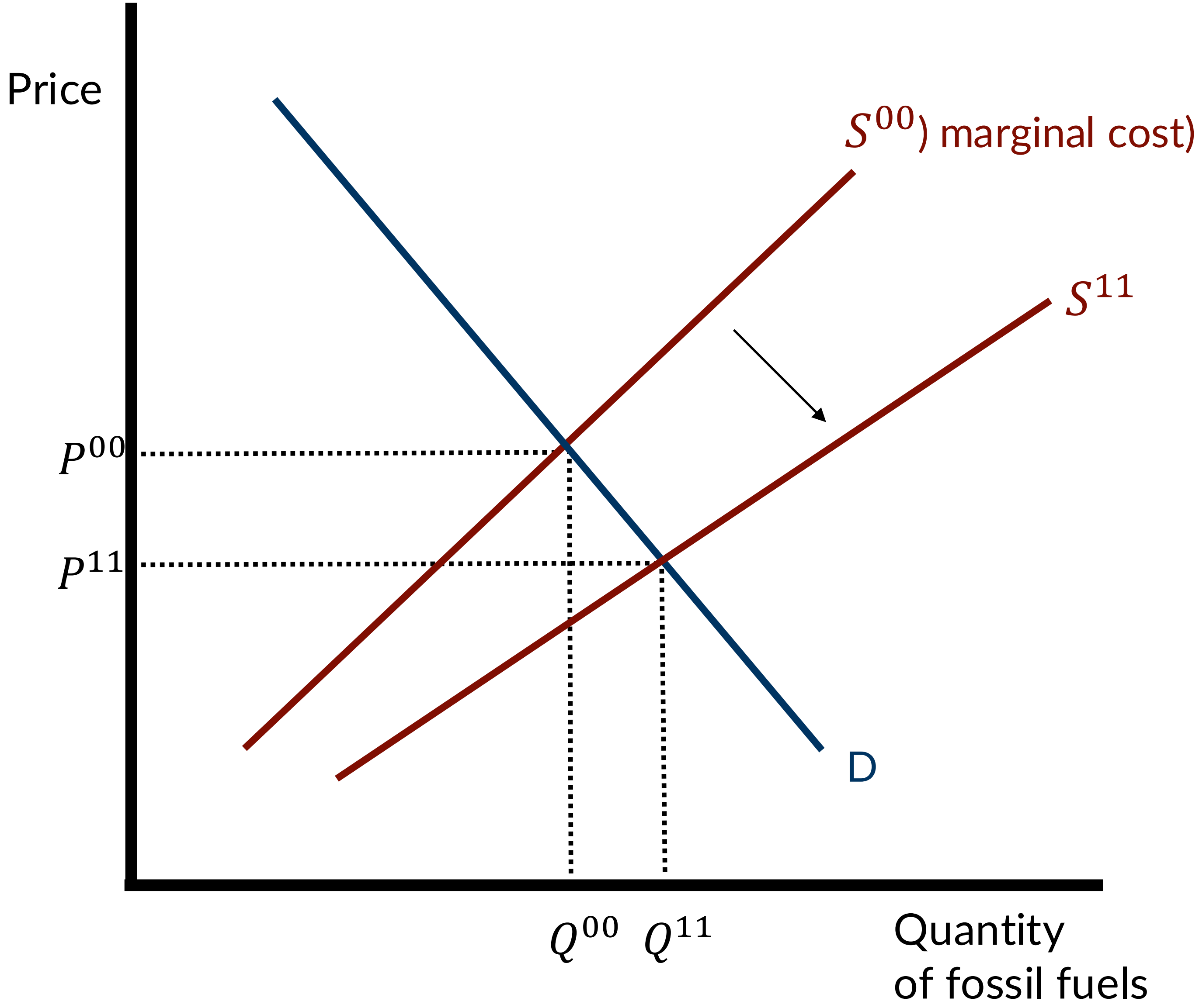
- **Will the world use less fossil fuel in 2050 than today?**
- **Supply theory**
  - Will we run out of cheap fossil fuels?
- **Demand theory**
  - Do cheaper alternatives displace fossil fuels?

A “supply theory” for getting off fossil fuels



- If fossil fuels become more scarce over time, that will reduce their supply ($S^{00} \rightarrow S^{11}$)
- This drives up the equilibrium price of fossil fuels, and alternative fuels become more “cost competitive”

A “supply theory" for getting off fossil fuels

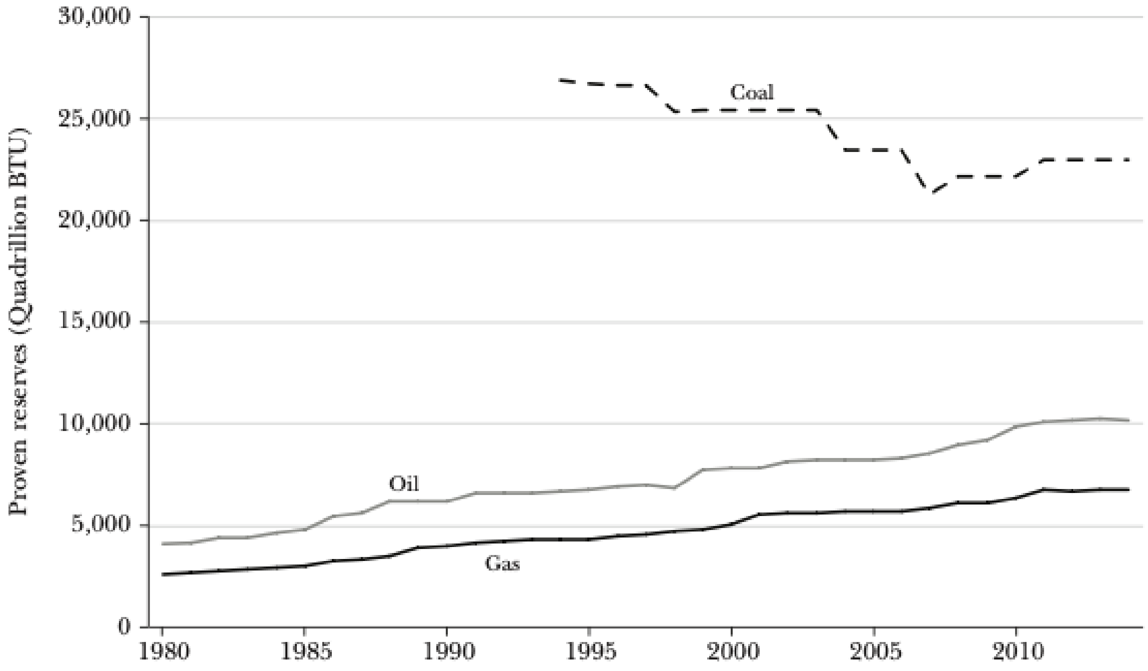


- Technology shifts supply out.
  - Fracking made shale oil and gas cheap to extract.
- Price falls, quantity rises. Scarcity recedes.

A “supply theory" for getting off fossil fuels

- *Proven reserves*: recoverable at current prices and technology.
- Reserves have grown for 40 years. Most countries find more oil and gas each year than they use.
- Scarcity alone will not end fossil fuels in the foreseeable future.
  - Coal may be the exception.

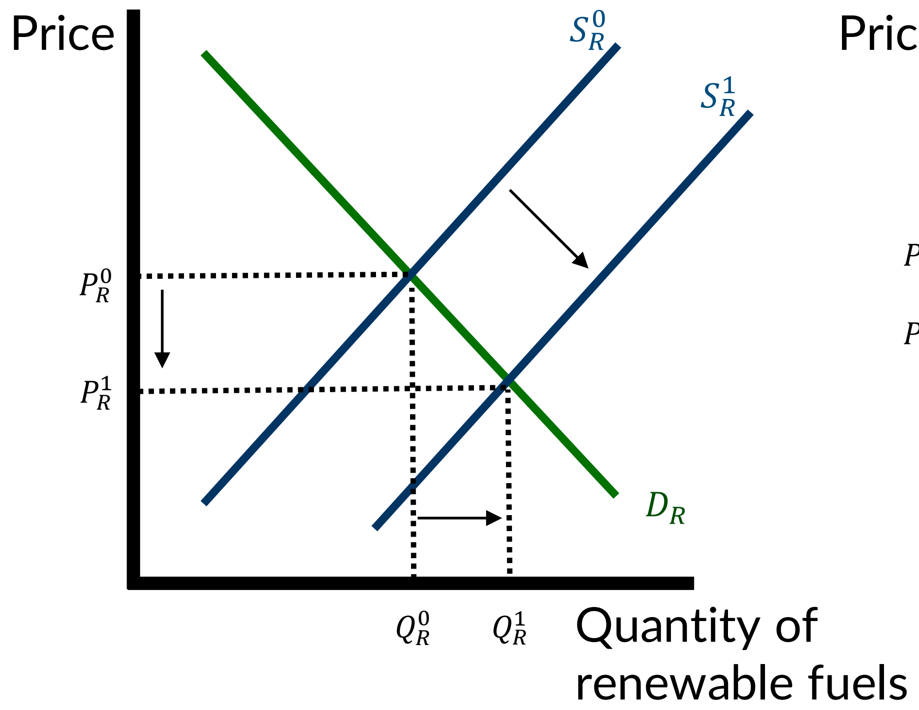
Figure 2
Proven Reserves of Oil, Natural Gas, and Coal over Time



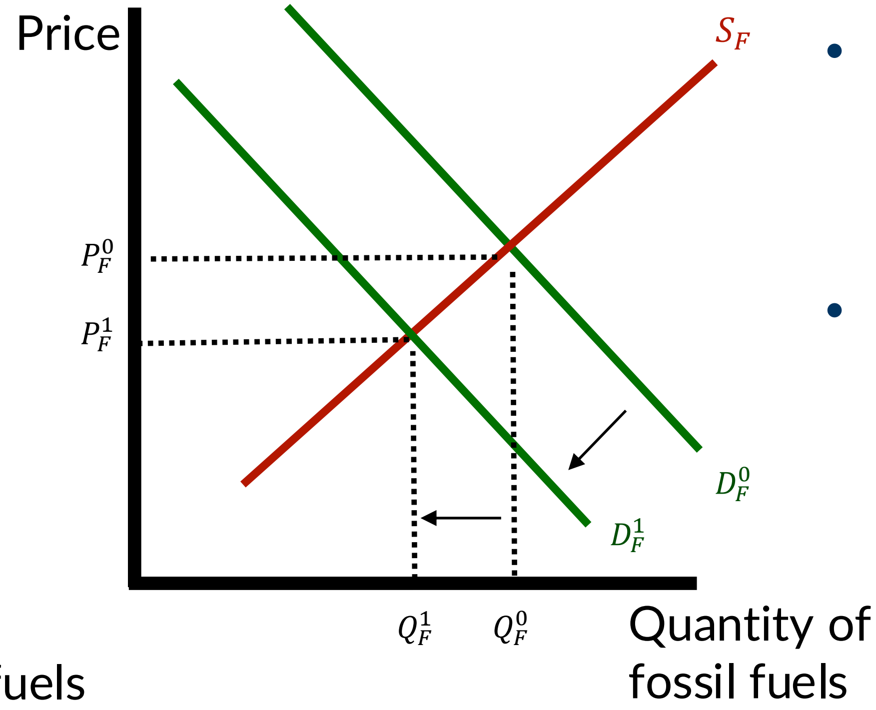
Source: BP Statistical Review of World Energy, 2015.
Note: The figure plots annual reserve estimates in absolute terms for oil, natural gas, and coal.

A "demand theory" for getting off fossil fuels

Market for renewable electricity generation



Market for fossil fuel electricity generation



- Consumers buy electricity, not fuel.
- Renewable and fossil generation are substitutes.
- Cheaper renewables:
  - renewable supply shifts out, price falls, quantity rises.
- Fossil fuel demand shifts in:
  - less fossil-fueled electricity, at a lower price.

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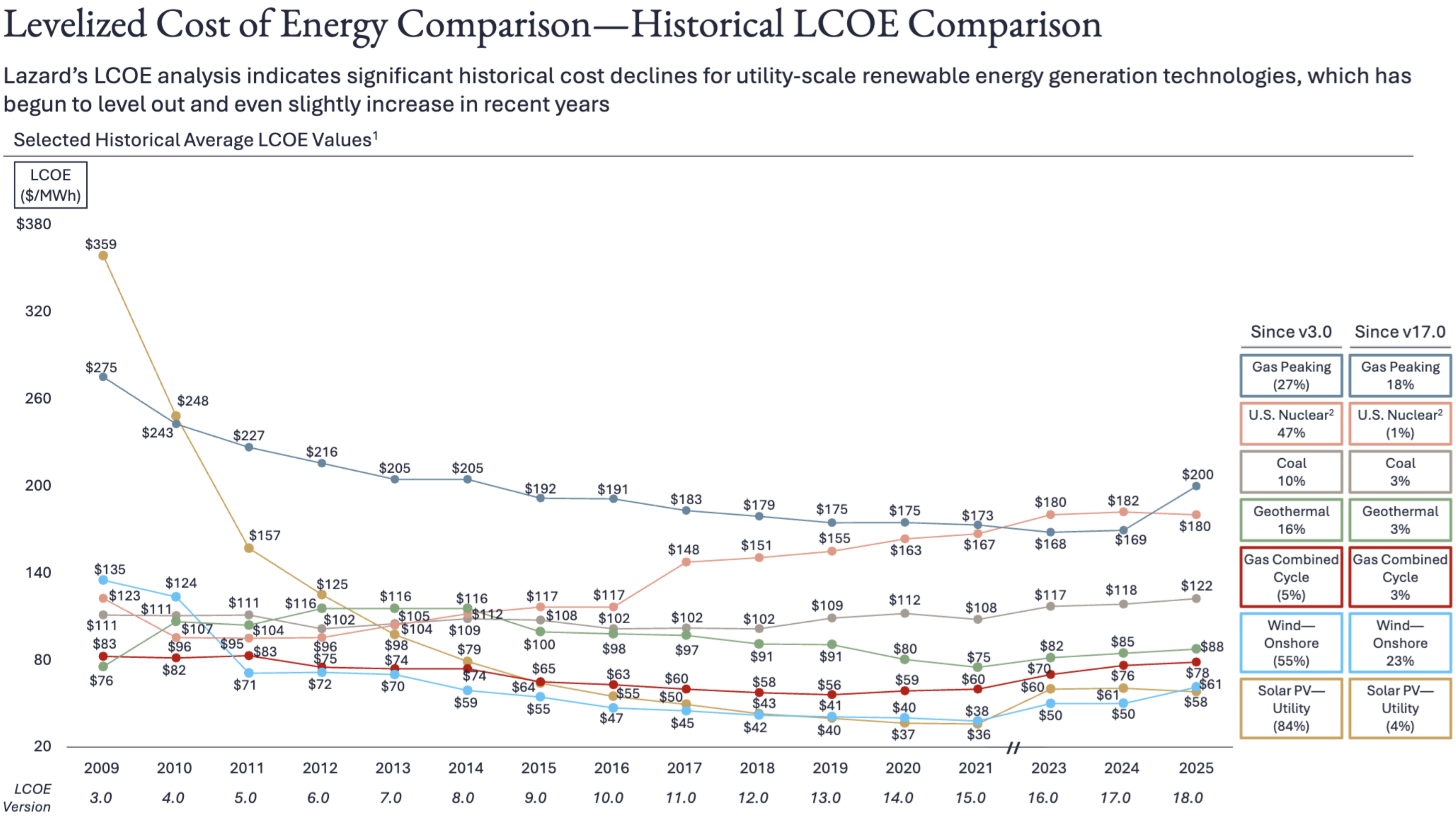
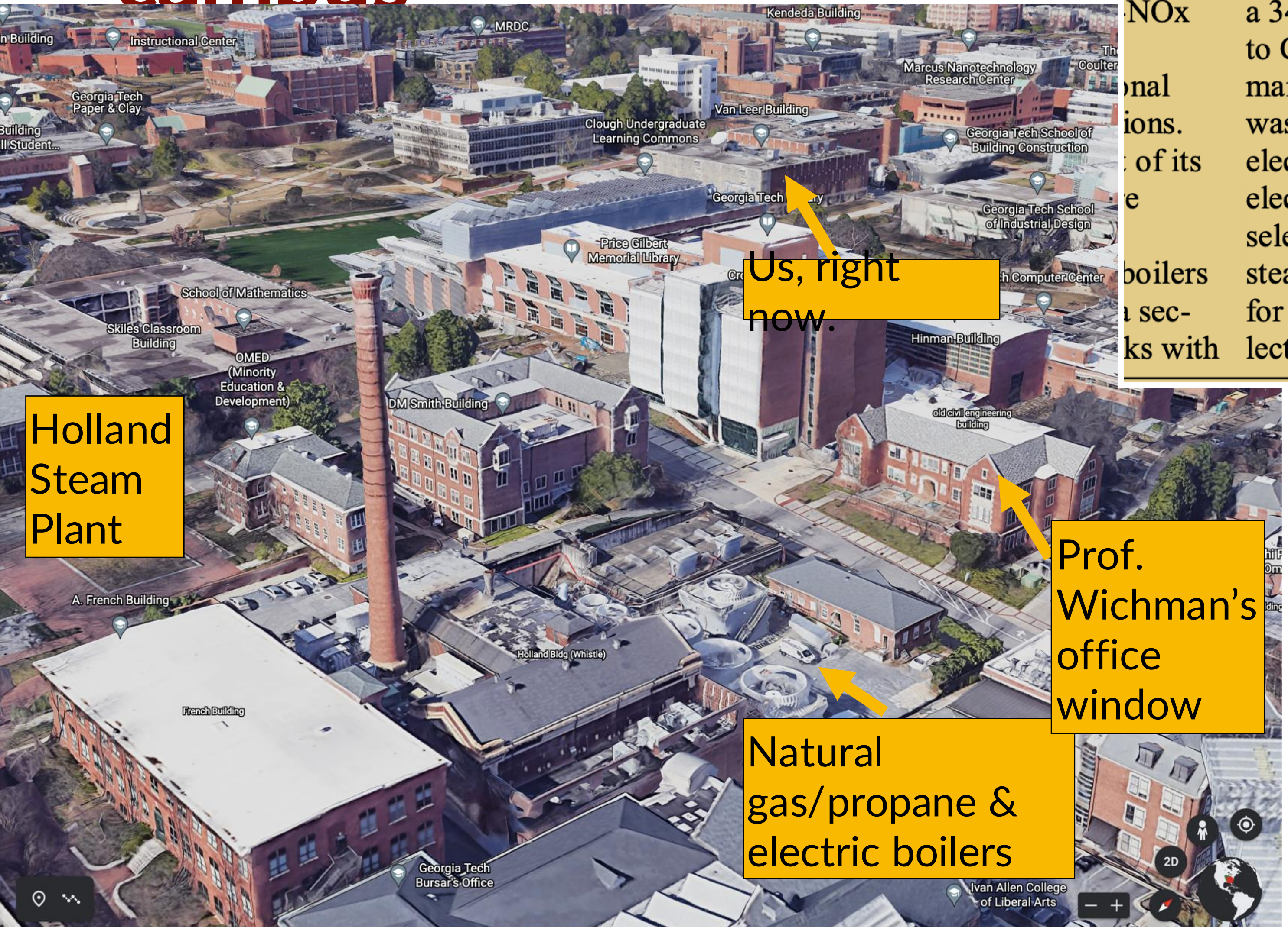
Cost of renewable energy is falling

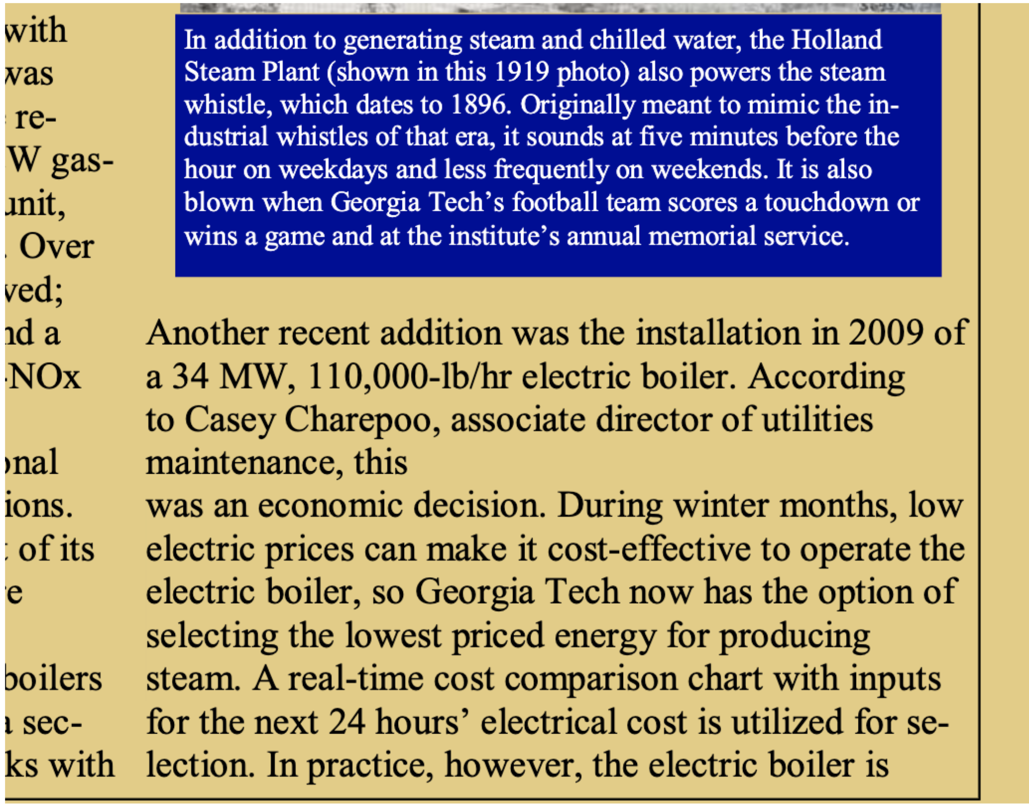
- **LCOE:** discounted lifetime cost per MWh, so up-front and operating costs compare on one scale.
- Wind and solar are now the cheapest new generation.
- Gas is close.
- Coal and nuclear are not.



Source:
https://www.lazard.com/media/eijnqja3/lazards-lcoeplus-june-2025.pdf

Cost-competitiveness and fossil fuels on campus





In addition to generating steam and chilled water, the Holland Steam Plant (shown in this 1919 photo) also powers the steam whistle, which dates to 1896. Originally meant to mimic the industrial whistles of that era, it sounds at five minutes before the hour on weekdays and less frequently on weekends. It is also blown when Georgia Tech's football team scores a touchdown or wins a game and at the institute's annual memorial service.

Another recent addition was the installation in 2009 of a 34 MW, 110,000-lb/hr electric boiler. According to Casey Charepoo, associate director of utilities maintenance, this was an economic decision. During winter months, low electric prices can make it cost-effective to operate the electric boiler, so Georgia Tech now has the option of selecting the lowest priced energy for producing steam. A real-time cost comparison chart with inputs for the next 24 hours' electrical cost is utilized for selection. In practice, however, the electric boiler is

Source: https://docplayer.net/18490916-Historic-georgia-tech-archibald-d-holland-central-heating-and-cooling-plant.html

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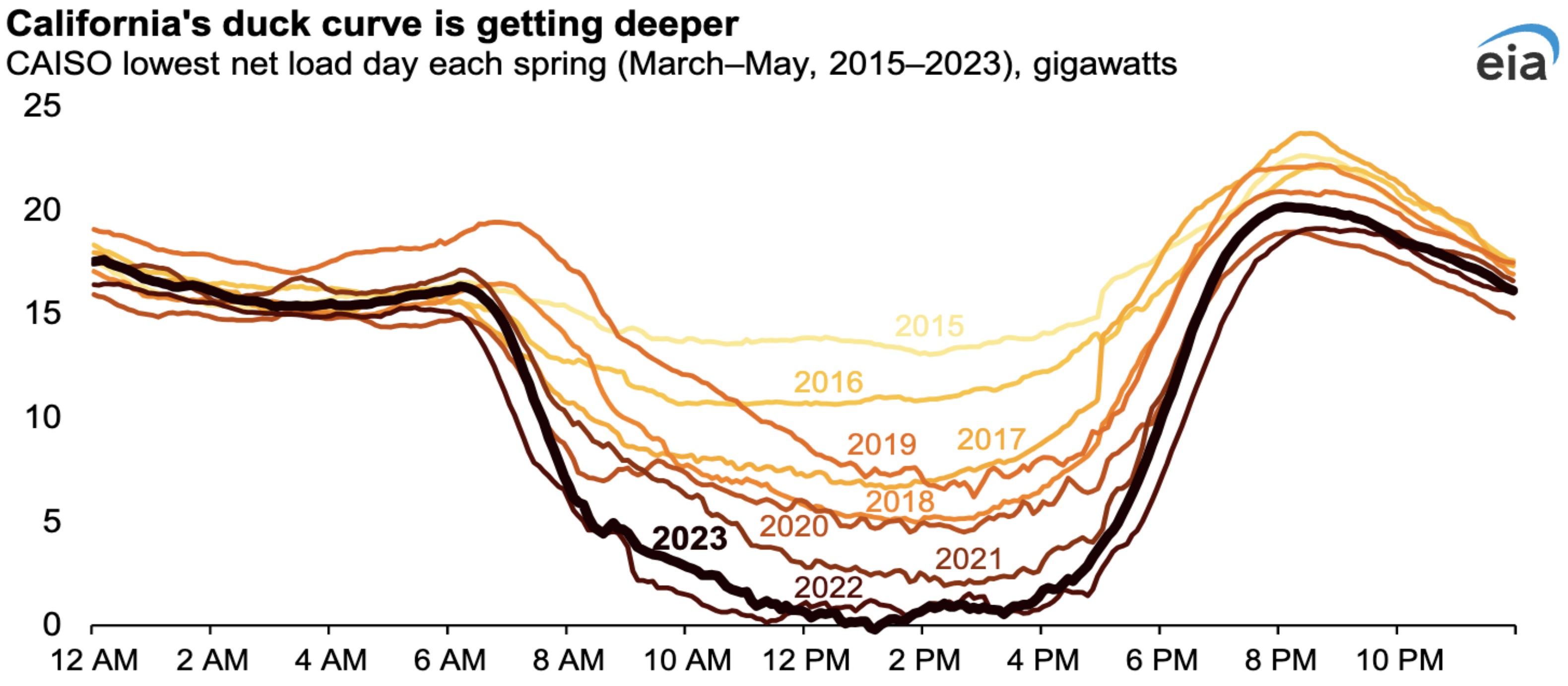
Renewables have a few challenges…

- **Intermittency.**
  - The sun sets and the wind stops.
- **Location.**
  - The best resources are far from people.
  - Transmission is slow and costly to build.
- **Storage.**
  - Grid batteries shift midday solar to the evening peak. Costs are falling fast.

JUNE 21, 2023

As solar capacity grows, duck curves are getting deeper in California



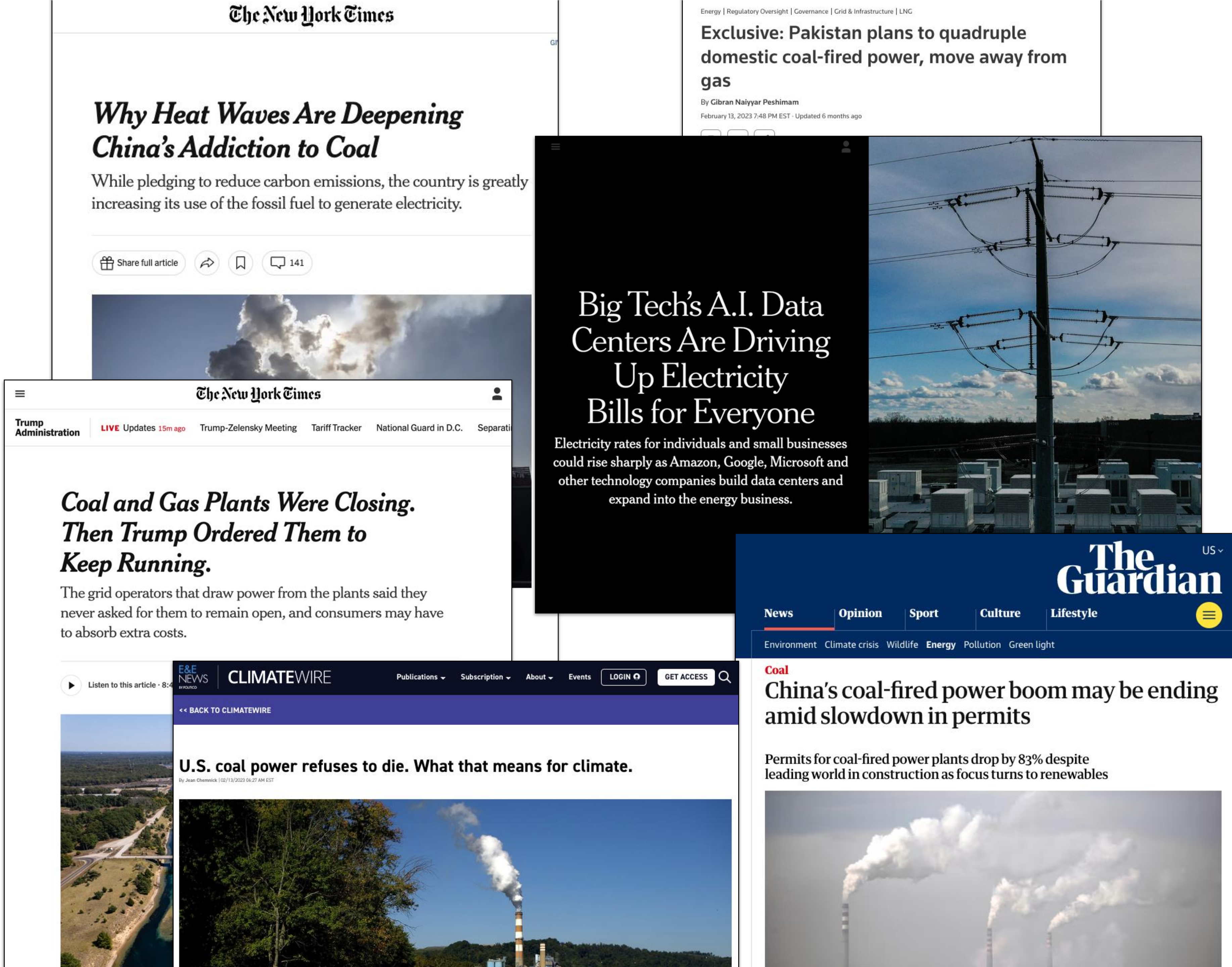
Data source: California Independent System Operator (CAISO)

- As solar penetration increases, lots of generation during the "sunny" parts of day.
- But, risks over generation (supply > demand).
- Leads to curtailing solar and reduces value of adding more renewables.
- "Ramping" (i.e., firing up fossil generators) in the evening is costly.

Source: https://www.eia.gov/todayinenergy/detail.php?id=56880

The end of fossil fuels for electricity generation doesn't seem likely based on market forces alone

- Coal plants (with lifespans of 40-50 years) are still being built.



The New York Times

Why Heat Waves Are Deepening China's Addiction to Coal

While pledging to reduce carbon emissions, the country is greatly increasing its use of the fossil fuel to generate electricity.

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Exclusive: Pakistan plans to quadruple domestic coal-fired power, move away from gas

By Gibran Naiyyar Peshimam

February 13, 2023 7:48 PM EST · Updated 6 months ago

Big Tech's A.I. Data Centers Are Driving Up Electricity Bills for Everyone

Electricity rates for individuals and small businesses could rise sharply as Amazon, Google, Microsoft and other technology companies build data centers and expand into the energy business.

The New York Times

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Coal and Gas Plants Were Closing. Then Trump Ordered Them to Keep Running.

The grid operators that draw power from the plants said they never asked for them to remain open, and consumers may have to absorb extra costs.

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The Guardian

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Coal

China's coal-fired power boom may be ending amid slowdown in permits

Permits for coal-fired power plants drop by 83% despite leading world in construction as focus turns to renewables

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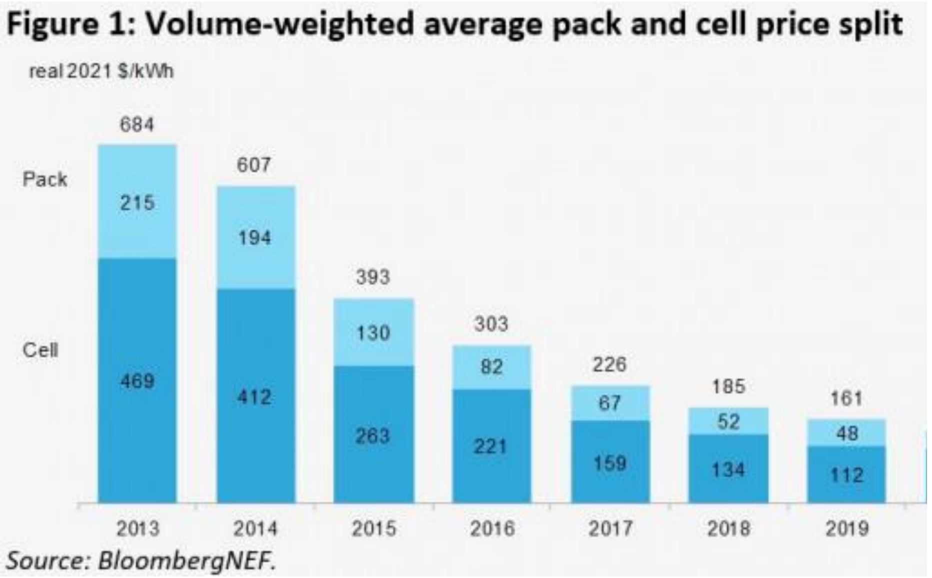
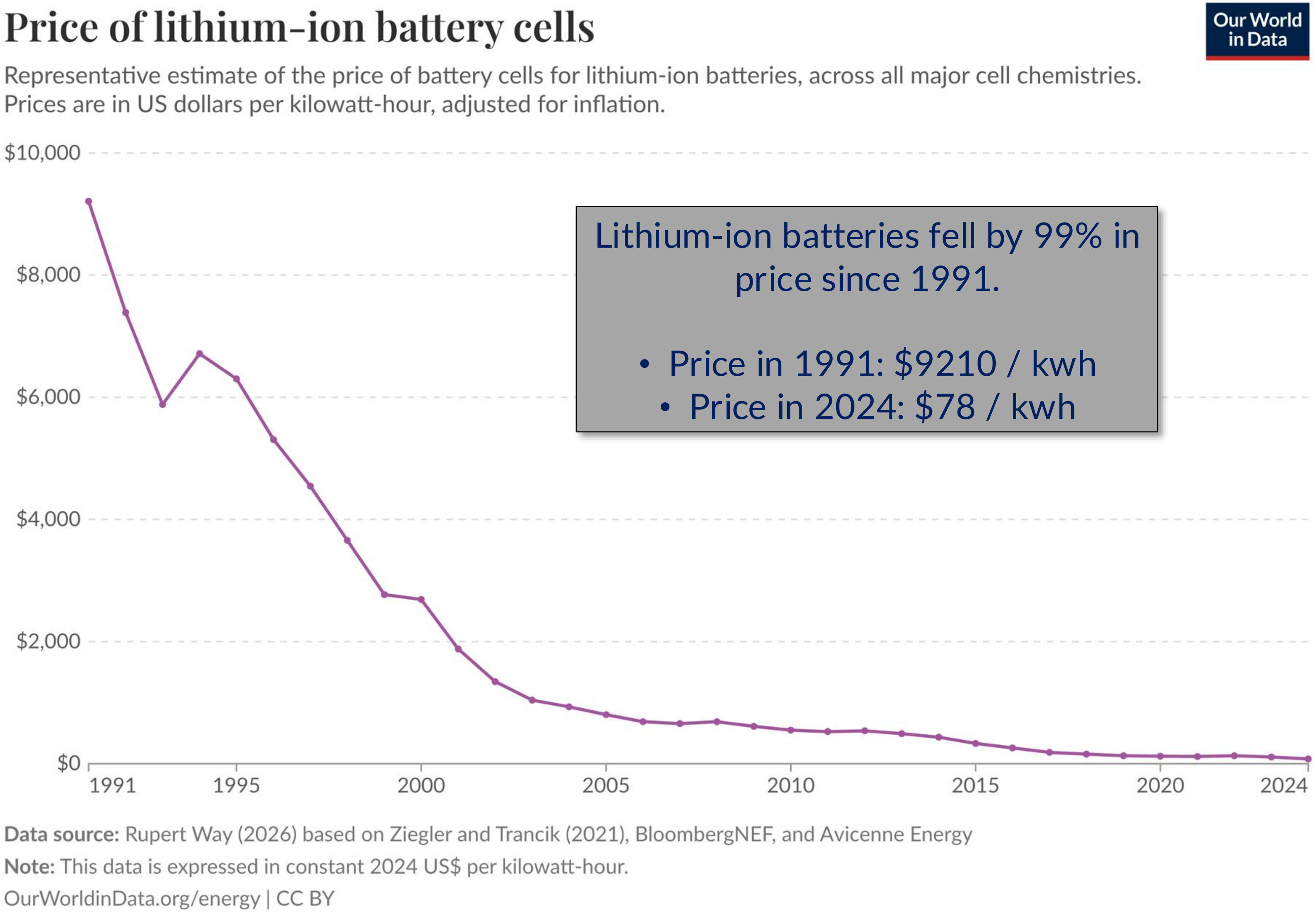
U.S. coal power refuses to die. What that means for climate.

Does the “demand theory” work getting cars off of oil?

- Technology has cut the cost of alternatives to oil.



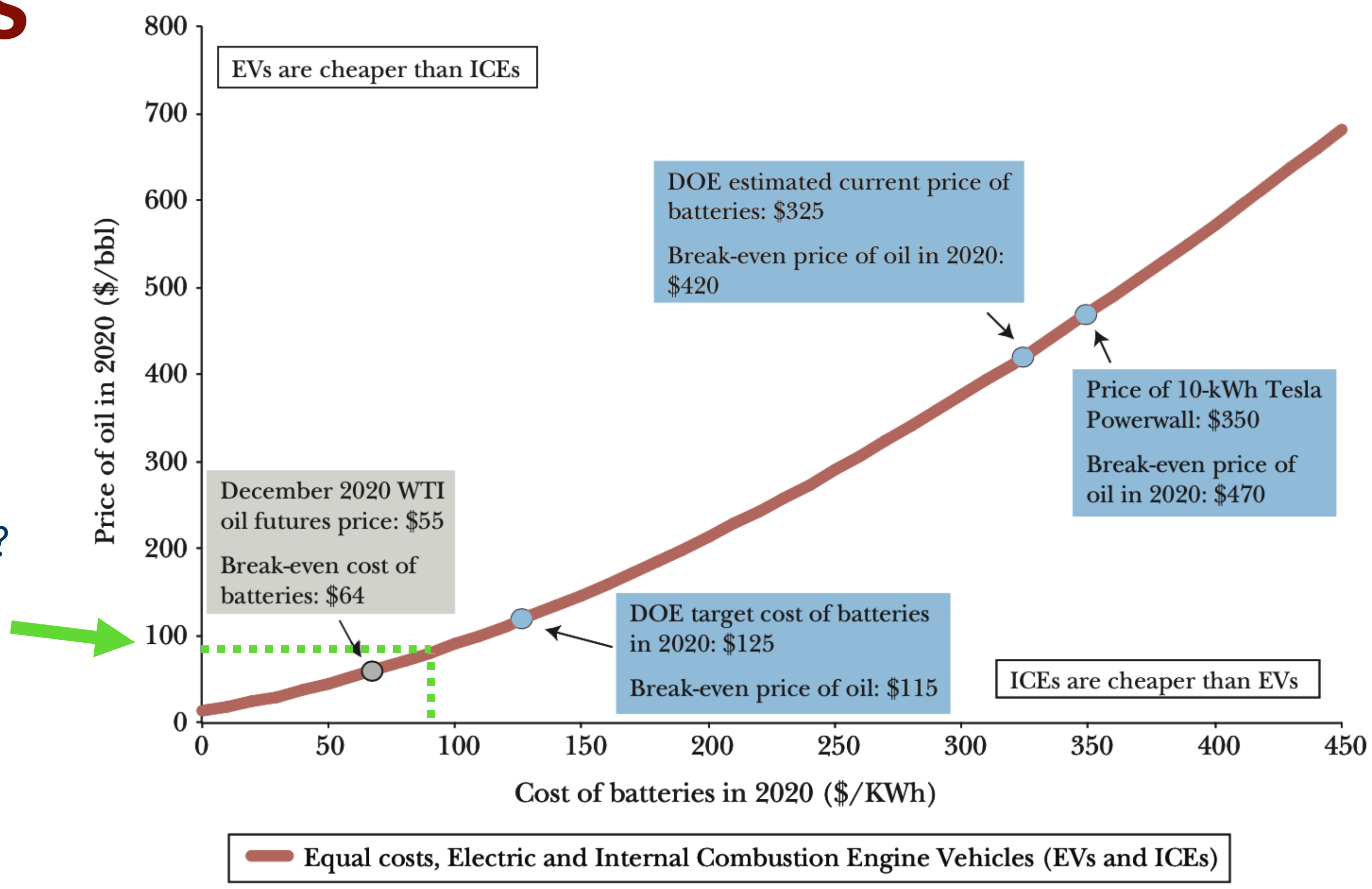


Comparing costs of cars and EVs

- Break-even battery price for a given oil price.
  - E.g., how much would EV battery packs have to be to be competitive with ICE vehicles at a given oil price?
- Oil (WTI):
  - ~$80/bbl (Aug 2026).
- Higher oil prices raise the break-even battery price.
- EV battery packs averaged $108/kWh in 2025 (BNEF).
  - EVs are close to parity on this metric.

Figure 6

Break-even Oil and Battery Costs



Notes: This graph plots the relationship between oil prices and battery costs such that the present discounted value of owning and operating an internal combustion engine vehicle equals the present discounted value of owning and operating an electric vehicle assuming the year is 2020. We assume the vehicle is driven 15,000 miles per year. We use the average retail price of electricity of 12.2 cents per kWh to charge the electric vehicle. We estimate the relationship between oil prices and gasoline prices using monthly data from 1993 to 2015 and assume a log–log relationship. We size the battery such that the electric vehicle has a range of 250 miles, assuming electricity consumption of 0.3 kWh per mile. The cost of the electric vehicle is reduced by $1,000 to reflect the lower costs associated with electric motors relative to the internal combustion engine vehicle. The fuel economy of internal combustion engines is assumed to grow at an annual rate of 2 percent, consistent with Knittel (2011).

Electric cars use electricity…

…electricity is (still) predominantly generated by fossil fuels in the US

- Charging on fossil power still emits CO2, but EVs are generally more efficient
- The effect depends on the marginal generator: which plant ramps when you plug in, where, and when.
- Electrifying vehicles lets future renewable growth count for transport too.

Table 2. Summary of External Cost Estimates (2021$)

External damages category	National average (per gallon)	National average (per mile)	Range across counties	Comparison with previous estimates
Carbon dioxide emissions	\$1.71	\$0.07	None	Well above previous estimates due to higher social cost of carbon (Rennert et al. 2022)
Local pollution	\$0.79	\$0.03	\$0.03–\$8.25 per gallon	Similar to previous estimates, e.g., Parry et al. (2007)
Congestion	\$0.27	\$0.01	\$0.00–\$0.12 per mile	Similar to previous national-level estimates, but we show substantial variation by county
Accidents	\$1.08	\$0.05	None	Estimate is from the literature; consistent with most other studies
Sum	\$3.85	\$0.16	\$2.83–\$13.44 per gallon \$0.12–\$0.56 per mile	Higher than most previous estimates, with wider range

Notes: Average pollution and congestion damages are weighted by county-level population. GHG damages are uniform across counties. Accident externalities are assumed to be equal across counties because of a lack of data.

Source: https://media.rff.org/documents/WP_23-09_qRXBRfb.pdf

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Not including the *external costs* of emitting GHGs is effectively the same as giving fossil fuels a massive subsidy.

“…in the absence of substantial greenhouse gas policies, the US and the global economy are unlikely to stop relying on fossil fuels as the primary source of energy.”

-Covert, Greenstone, Knittel (2016)

Key takeaways

- Technology cuts both ways:
  - cheaper fossil fuels and cheaper renewables.
- Wind and solar are the cheapest new generation, but imperfect substitutes (intermittency, location, storage).
- Coal-to-gas switching cuts CO2 but does not end fossil fuels.
- New electricity demand (data centers) is reviving nuclear and delaying coal retirements.
- Pricing externalities, CO2 and otherwise, is required for big reductions in fossil fuel use.

For next class

Why is climate change an economic problem?
Read: Keohane & Olmstead, Chs. 2, 4 & 5

www.caseyjwichman.com/econ4210#week2